

HIGH PERFORMANCE SRAM DESIGN FOR ENERGY EFFICIENT SYSTEMS

S13APROJ2 – PROJECT WORK (PHASE – II)

Submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Electronics and Communication Engineering

by

N UMESH SURYA KIRAN (Reg No. 42130311)

VADDI UGANDHAR (Reg No. 42130516)



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SCHOOL OF ELECTRICAL AND ELECTRONICS**

SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)**

CATEGORY - 1 UNIVERSITY BY UGC

**Accredited with Grade "A++" by NAAC | Approved by AICTE
JEPPIAAR NAGAR, RAJIV GANDHI SALAI, CHENNAI - 600 119**

APRIL – 2026



SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CATEGORY - 1 UNIVERSITY BY UGC

Accredited with 'A++' grade by NAAC | Approved by AICTE

www.sathyabama.ac.in

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that this Project Report is the Bonafide work of
VADDI UGANDHAR (42130516), N U SURYA KIRAN (42130311) who carried
out project entitled **High Performance SRAM Design for Energy Efficient
Systems** under my guidance from September 2025 to April 2026.

Internal Guide & Head of Department

Dr. T. RAVI, M.E., Ph.D.,

Submitted for Viva voce Examination held on 15.04.26

Internal Examiner

External Examiner

DECLARATION

We **N UMESH SURYA KIRAN (42130311), VADDI UGANDHAR (42130516)** hereby declare that the project entitled **High Performance SRAM Design for Energy Efficient Systems** done by us under the guidance of **Dr. T. RAVI, M.E., Ph.D.**, at **SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Electronics and Communication Engineering.

DATE:

PLACE:

SIGNATURE OF CANDIDATES

1.

2.

ACKNOWLEDGEMENT

We are pleased to acknowledge my sincere thanks to the **Board of Management** of **SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY** for their kind encouragement in doing this project and for completing it successfully. We are grateful to them.

We convey our thanks to **Dr.T.RAVI M.E., Ph.D.**, Head of the Department, Department of Electronics and Communication Engineering for providing me the necessary support and details at the right time during the progressive reviews.

We would like to express my sincere and deep sense of gratitude to my Project Guide **Dr.T.RAVI M.E., Ph.D.**, for his valuable guidance, suggestions and constant encouragement that paved way for the successful completion of my project work.

We wish to express our thanks to all Teaching and Non-Teaching staff members of the Department of Electronics and Communication Engineering who were helpful in many ways for the completion of the project.

ABSTRACT

Static Random Access Memory (SRAM) plays a critical role in modern integrated circuits, particularly in applications requiring high-speed and low-power operation such as Internet-of-Things (IoT) devices, portable electronics, and edge computing systems. With the increasing demand for energy-efficient designs, aggressive supply voltage scaling has become a widely adopted technique to reduce power consumption. However, voltage scaling significantly impacts SRAM performance by degrading noise margins, increasing susceptibility to process variations, and reducing reliability during read and write operations. Conventional SRAM designs face a fundamental trade-off between stability, power consumption, and area, making it challenging to achieve optimal performance under low-voltage conditions. In this work, a custom optimized 6T SRAM cell is proposed to address these challenges through intrinsic circuit-level optimization. The design focuses on improving read stability, write ability, and reducing leakage power by carefully optimizing transistor sizing and balancing the strengths of pull-up, pull-down, and access transistors. Unlike conventional approaches that rely on assist techniques or additional circuitry, the proposed design enhances performance within the standard 6T structure, thereby maintaining area efficiency and simplicity. The proposed 6T Architecture cell is designed and simulated using Cadence Virtuoso and evaluated under low-voltage operating conditions. Key performance parameters such as power consumption, read delay, write delay, and static noise margin (SNM) are analyzed and compared with an existing 7T SRAM design. Simulation results demonstrate that the proposed design achieves significant reduction in power consumption and leakage current while maintaining comparable read performance and improved write speed. The optimized design also exhibits sufficient noise margin, ensuring reliable operation under reduced supply voltage conditions.

TABLE OF CONTENTS

Chapter No	Description	Page No
	ABSTRACT	v
	LIST OF FIGURES	viii
	LIST OF TABLES	ix
1	INTRODUCTION	1
2	LITERATURE SURVEY	6
3	EXISTING METHODOLOGY	18
	3.1 7T SRAM Architecture	19
	3.2 Scope	19
	3.3 Requirement Analysis	21
	3.3.1 Functional Requirements	21
	3.3.2 Performance Requirements	21
	3.3.3 Area and Integration Requirements	21
	3.3.4 Tool and System Requirements	22
	3.4 Design Specifications	22
	3.5 Constrains and Assumptions	23
	3.6 Risk Analysis	24
4	PROPOSED LOW-POWER OPTIMIZED 6T SRAM ARCHITECTURE AND IMPLEMENTATION	26
	4.1 Proposed Architecture	26
	4.2 Transistor – Level Design and Optimization	28
	4.3 Operation of proposed SRAM Cell	28
	4.3.1 Hold Operation	28
	4.3.2 Read Operation	29
	4.3.3 Write Operation	29
	4.4 SRAM Macro Architecture and Integration	30
	4.4.1 Memory Array Organization	30
	4.4.2 Row Decoder and Wordline Driver	31

Chapter No	Description	Page No
	4.4.3 Bitline Architecture	32
	4.5 Implementation using Cadance Virtuoso	33
	4.6 Testing and Validation plan	34
	4.7 Result and Discussion	35
	4.7.1 Transient Analysis and Functional Verification	35
	4.7.2 Static Noise Margin (SNM) Analysis	37
	4.7.3 Read Delay Analysis	38
	4.7.4 Write Delay Analysis	39
	4.7.5 Power Consumption Analysis	39
	4.7.6 Energy Per Operation Analysis	41
	4.7.7 Overall Performance Summary	41
5	CONCLUSION	43
	REFERENCES	44
	ANNEXURES	47

LIST OF FIGURES

Figure No	Description	Page No
1.1	Typical SRAM Cell Structure	2
1.2	Butterfly Curve for SNM Measurement	3
3.1	Existing 128×64 SRAM macro with 7T SRAM cell	23
4.1	Conventional 6T SRAM Cell Architecture	27
4.2	Read & write Operation Waveform of the Proposed 6T SRAM Cell	29
4.3	SRAM Macro Architecture Block Diagram	30
4.4	4-Bit Addressed SRAM Array Organization	31
4.5	Wordline Driver Circuit	31
4.6	Decoder Circuit	32
4.7	Control and Timing Signal Sequence	33
4.8	Cadence Virtuoso Schematic of the 6T SRAM Cell	34
4.9	Transient Analysis – 6T SRAM	35
4.10	Transient Analysis – 7T SRAM	36
4.11	6T SRAM Macro Output Waveforms	36
4.12	7T SRAM Macro Output Waveforms	37
4.13	DC Analysis – 6T SRAM	37
4.14	DC Analysis – 7T SRAM	38
4.15	Power Analysis – 6T SRAM	39
4.16	Power Analysis – 7T SRAM	40
4.17	Power Analysis – 6T Architecture	40
4.18	Power Analysis – 7T Architecture	40

LIST OF TABLES

Table No	Description	Page No
4.1	Static Noise Margin Comparison	38
4.2	Read Delay Comparison	38
4.3	Write Delay Comparison	39
4.4	Power Comparison	41
4.5	Energy Comparison	41
4.6	Performance Comparison of 6T and 7T Architecture	42